SOURAV ROY

Mathematics & Algorithms → Quantitative Research & Data Science

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PROFILE

My background combines two separate TIFR master's degrees: most recently an M.Sc. in Computer and System Sciences (2021–2026) from TIFR STCS, Mumbai, preceded by an M.Sc. in Mathematics (2019–2021) from TIFR-CAM, Bengaluru. I study probability, randomized algorithms, convex geometry, computational mathematics and optimization, and develop reproducible Python/C++ experiments with explicit numerical limitations. I apply this mathematical and computational foundation to data science, financial time series, forecasting, portfolio analysis and transaction-risk modelling. My work emphasizes careful assumptions, chronological evaluation, leakage control, reproducibility and evidence-based interpretation. I am transitioning toward quantitative research, data science and risk analytics, including quantitative finance.

TECHNICAL SKILLS

Programming & data: Python, C++17, NumPy, pandas, SciPy, scikit-learn, XGBoost, Matplotlib, CVXPY, Clarabel, SQL, SQLite; C, MATLAB, SageMath, Mathematica.

Quantitative & statistical methods: Probability, PCA, Ridge regression, Random Forest, KMeans, Gaussian HMMs, covariance estimation, minimum-variance optimization, risk forecasting, convex optimization, randomized algorithms and numerical simulation.

Research & evaluation: Chronological validation, precommitted holdout evaluation, fixed-cutoff forecasting, paired ranking and review-budget analysis, model calibration, feature ablation, backtesting with transaction costs, leakage auditing, forecast freezing, sampling, complexity analysis, numerical invariants, reproducibility checks, Python unittest and pytest.

Tools: Git, GitHub, JupyterLab / Jupyter Notebook, VS Code, conda / Anaconda, macOS Terminal, DB Browser for SQLite, sqlite3 CLI, Homebrew, Apple Clang (C++17), LaTeX. These reflect tools I have used, with different levels of proficiency.

DATA SCIENCE, QUANTITATIVE FINANCE & RISK PROJECTS

Quant AI Market Lab | Independent quantitative research | completed v1.0.0 | 2026 | **Repository**

- Built a validated eight-ETF workflow covering returns, covariance, PCA, KMeans and Gaussian HMM market-condition models, Ridge and Random Forest forecasts of next-five-day SPY return RMS, and long-only minimum-variance portfolios.
- Verified return identities, covariance symmetry/PSD properties, eigendecomposition and explained variance. The first three PCs explained 79.01% of standardized historical return variation; on 565 common chronological dates Ridge MAE was 0.003201 versus 0.003443 for Random Forest.
- Compared five covariance methods, three transaction-cost assumptions, nine benchmark scenarios and 32 strategy-period robustness checks. Minimum-variance approaches reduced realized volatility in reported subperiods but underperformed SPY buy-and-hold on raw return; May-August 2026 are described as pseudo-out-of-sample rather than an untouched holdout.
- Compared 2-6 Gaussian HMM states, diagonal/full covariance options, multiple initializations, likelihood, AIC/BIC, occupancy and chronological diagnostics; retained an interpretable three-state full-covariance specification without claiming economically proven regimes. Examined Ridge coefficient stability across 87 monthly model refits.

Quant Market Lab II - Signal Processing & Prospective Forecasting | v0.1 baseline frozen; extensions in progress | 2026 | **Repository**

- Fixed the information cutoff at 31 August 2026; implemented Ridge, constant and rolling 20-day RMS baselines at four lead times, preserved 12 versioned September-December forecasts, and recorded model/data/source fingerprints with target-timing and prefix-invariance tests.
- Across 87 historical zero-gap windows Ridge had the lowest average absolute error in the repository comparison. September remains a retrospective check; nine October-December five-session forecasts were published from the frozen August-cutoff CSV without refitting. I implemented EWMA with separately frozen August/October information sets, then reproduced signed, squared and absolute-return autocorrelation without changing the forecasts. November and December outcomes remain pending; autoregressive extensions remain future work.
- Evaluated the first completed October five-session window: realized SPY daily-return RMS was 0.5263%, compared with the frozen rolling-20 forecast of 0.5780%, Ridge forecast of 0.8118% and constant baseline of 0.8791%. This is one evaluation window, not proof of predictive superiority.
- Reproduced August-data lag-one autocorrelations of -0.116 (signed), 0.420 (squared) and 0.352 (absolute returns). Verified the preserved original adjusted-price CSV against its SHA-256 manifest and documented differences between later download vintages. Kept reconstructed EWMA experiments separate from the original frozen forecasts.

Payment Guard | Completed synthetic fraud-review ranking study | 2026 | **Repository** · **Evidence and limits**

- Audited 1,090,000 synthetic AMLNet v2 transactions, reconciled 1,411 positive labels, inspected nested metadata and potential target leakage, prepared chronological splits, verified a SQLite import and investigated training-period patterns using SQL.
- Implemented regularized logistic regression and XGBoost with pandas/NumPy, scikit-learn preprocessing, chronological training and model selection, sigmoid/isotonic calibration, feature ablation and paired review-budget analysis. Frozen March–April 2026 final holdout: 253,190 transactions, 278 fraud labels; at the precommitted top-1% review capacity (2,532 reviews/model), XGBoost captured 267 positives versus logistic regression's 254 (+13 captured labels). Logistic captured 225 versus 222 at the smaller 0.1% budget; superiority was conditional on the chosen objective and workload.
- Analyzed feature sensitivity: XGBoost at 1% captured 169 positives without `category` and 141 without both `category` and `type` (267 with all features). Decision-time provenance of these synthetic columns remains unverified. Published aggregate-only plots and metrics, freeze and hash checks, and an auditable evaluation; **186 tests passed locally**. This is a completed synthetic-data research milestone, not a deployed bank fraud-detection system or established financial savings.

ResearchAudit | Evidence-based validation project | rule-based baseline published | 2026 | **Repository**

- Defined an evidence-based multi-agent audit plan and froze Quant AI Market Lab v1.0.0 as the first target. I published an AST-based screening baseline: 34 Python files, 10 triaged findings and 52 passing test methods. Six claims retain provisional inspection support. Claim 7 is supported for tested cases with two input-validation gaps, recorded by characterization tests. Full numerical reproduction, independent detection-accuracy evaluation and multi-agent experiments remain open.
- Performed isolated transaction-cost function checks for Claim 7, supporting the tested cases while identifying two input-validation gaps; these checks do not reproduce the complete backtest. Human triage of 10 flagged locations did not confirm look-ahead errors at those locations, but this does not certify the full project.

MATHEMATICAL & COMPUTATIONAL RESEARCH

Approximating Mixed Volumes to Arbitrary Accuracy | 2025 | Hariharan Narayanan and Sourav Roy | **arXiv paper**

- Co-authored a randomized approximation algorithm for mixed volumes of convex polytopes with integer vertices, combining convex optimization, Lorentzian polynomials and polytope subdivision; established multiplicative-error and failure-probability guarantees with polynomial runtime for a fixed number of distinct input polytopes under the stated assumptions.

Master's Thesis: Approximation of Mixed Volumes of Special Convex Bodies | TIFR Mumbai | March 2026 | Supervisor: Prof. Hariharan Narayanan | **Thesis PDF**

- Studied mixed volumes of a polytope and Euclidean ball using Minkowski sums, the Steiner polynomial, geometric decomposition, sampling and log-concavity; developed and analysed noisy binary search for a scaling parameter, including decision-error and runtime bounds, while distinguishing the idealized oracle model from practical approximate optimization.
- Implemented random-walk samplers for estimating volumes of convex bodies, including ball walk and hit-and-run variants, together with geometric membership/decomposition procedures and mixed-volume computations.

Exploring the Efficient Computation of Permanents of Graph Laplacians | TIFR | December 2022 | Supervisor: Prof. Piyush Srivastava | **Project report PDF**

- Proved the permanent analogue of Brenner's result for diagonally dominant matrices; explored tensor-product formulations, recursive expressions for paths, cycles and trees, and Pfaffian methods.

RSA Encryption and Decryption in Python | INSPIRE Summer Project, Presidency University Kolkata | 2019; revised 2026 | **Repository**

- Implemented educational RSA key generation and encryption/decryption using Miller-Rabin primality testing, the extended Euclidean algorithm and modular exponentiation. The 2026 revision adds UTF-8/ input validation and seven unit tests and documents correctness and public-key lookup attack limits of byte-wise textbook RSA.

Mixed Volume Approximations | Python and C++ numerical experiments | October 2026 | Repository

- I implemented numerical versions of the framework in my master's thesis, Approximation of Mixed Volumes of Special Convex Bodies (Chapters 4–6): B1 estimates volume, B2 samples a geometric component label, and B3 estimates component probabilities. Algorithms 1–3 organize sampling, scaling-parameter search and mixed-volume estimation. I checked the cube-plus-ball estimates against exact formulas and ran matched Python/C++ pilot comparisons. Finite burn-in, correlated samples and floating-point calculations do not establish the ideal-oracle guarantees or certify mixing and accuracy.
- Compared matched Python/C++ pilot calculations across dimensions 7, 9 and 11 and scale parameters 0.5, 1.5 and 3.0; measured execution time and numerical error against exact cube cases, with reproducibility settings and limitations recorded in the repository.

Experiments on Random Walks | Computational convex geometry | Python/C++17 | October 2026 | **Repository · Interactive research website**

- Implemented Ball Walk, Dikin Walk, Coordinate Hit-and-Run, Random Hit-and-Run and a cube-specialized John's Walk on high-dimensional cubes, with readable explicit and optimized Python/C++17 code. Published source-labelled convergence checkpoints and cost comparisons for dimensions 1–10, with separate dimension-20 experiments. Distinguished empirical marginal/pairwise diagnostics from rigorous total-variation mixing; a browser JavaScript implementation allows exploratory runs but is not a native-language benchmark.

EDUCATION

M.Sc., Computer and System Sciences | Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai | 2021-2026 | First Class

M.Sc., Mathematics | Tata Institute of Fundamental Research, Centre for Applicable Mathematics, Bengaluru | 2019-2021 | First Class

B.Sc. (Honours), Mathematics | Serampore College, University of Calcutta | 2015-2018 | First Class

SELECTED COURSEWORK

Computer and System Sciences: Mathematical Foundations for Systems Sciences; Algorithms; Mathematical Logic; Cryptography; Advanced Probability; Polynomials and Applications; Topics in the Study of Markov Chains (half-course); Coding Theory; Topics in High Dimensional Geometry. Audited: Convex Optimization; Random Matrix Theory.

Mathematics: Linear Algebra; Algebra; Probability; Numerical Analysis; Real and Complex Analysis; Measure and Integration; Functional Analysis; Topology; Ordinary and Partial Differential Equations.

TIFR-CAM elementary PDE syllabus: method of characteristics; classification of second-order PDEs; Laplace, heat and wave equations; Green functions; maximum principles; separation of variables; and boundary-value solution methods. This is academic coursework, not a claim of professional derivatives-pricing experience.

PRESENTATIONS, ACADEMIC ACTIVITIES & RECOGNITION

- I spoke at the TIFR STCS volume-approximation seminar on 26 July 2024 and organized the Private Sequential Stateless Protocols seminar on 6 December 2024.
- Research presentation: Approximation of Mixed Volumes of Convex Bodies, STCS Symposium, TIFR, August 2025; seminar organization: Catalytic Graph Algorithms, STCS student seminar, TIFR, March 2026.
- Convex and Discrete Geometry Summer School, Alfréd Rényi Institute, Budapest, August-September 2023; School on Cluster Algebras, ICTS-TIFR, December 2018; Introduction to SageMath, ICT Mumbai and MTTS Trust, May-June 2020.
- TIFR / Department of Atomic Energy Research Scholarship, 2019-2026; DST INSPIRE Scholarship, 2015-2019.
- Qualified JAM Mathematics (2018) and Mathematical Statistics (2019); ISI M.Math admission test (2018); CMI M.Sc. Data Science written test (2019).
- Languages: Bengali, English and Hindi. References: three TIFR faculty referees available on request.